

GEOG 281A

Uncertainty

Figuring out how things relate to one another, formally

Dr. Peter Kedron



UNCERTAINTY

Couclelis (2003)

Intrinsic Uncertainty - Things that cannot be fully known

Undecidability	Things that cannot be proven false from within the formulation of a system
Complexity	Emergent properties from non-linear dynamics that are unpredictable
Vagueness	The challenge of dealing with categories and change (Sortes paradox, The Ship of Theseus)

Goodchild (2020)

Reinforcing that the map is not the territory

Measurements are never perfect: instruments have inherent limitations

Categories cannot be strictly replicated: interpretation and labeling of geographic entities are often difficult to pin down

Geospatial data are scale dependent: simplification and representation decisions depend on the scale selected for analysis

Maps appear too certain: maps looks stable and authoritative, they are not.

Uncertainty and error are spatially structured: Uncertainty is not homogeneous

Uncertainty propagates during analysis: The challenge is in the covariance

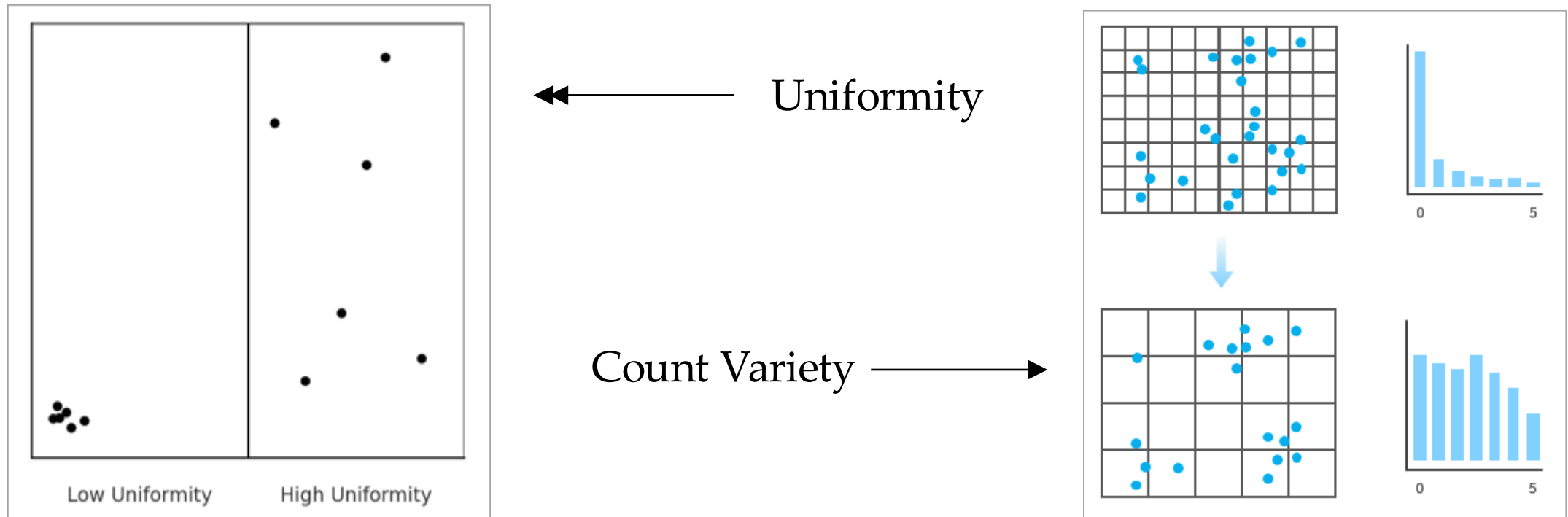
STUDENTS PERSPECTIVE

Do we need to develop better methods for incorporating uncertainty into our geospatial analyses so there is motivation to develop better systems for tracking uncertainty in data? Or do we need to first have standardized metrics of uncertainty in order to develop these methods of handling uncertainty? **Who Should be leading this movement?**

FROM RESEARCH TO PRACTICE

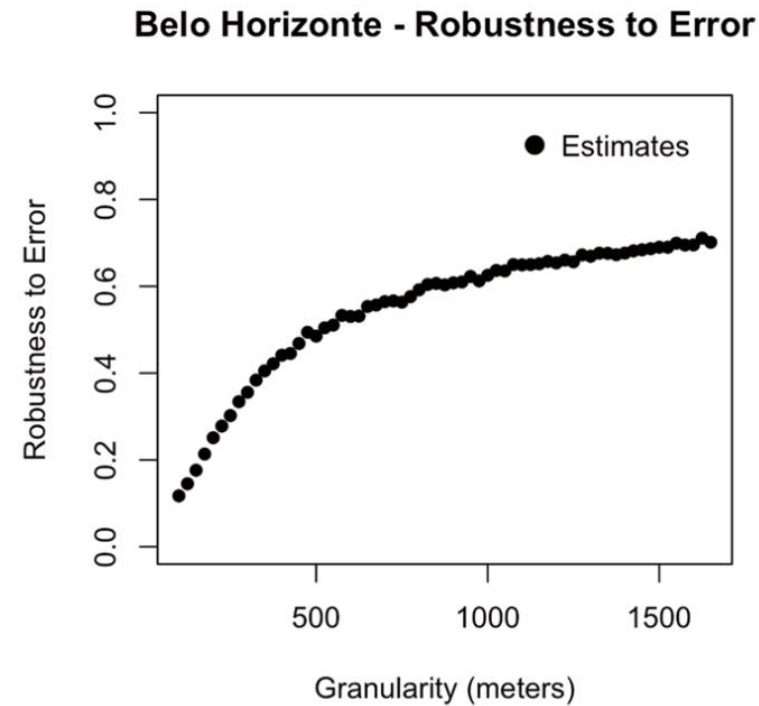
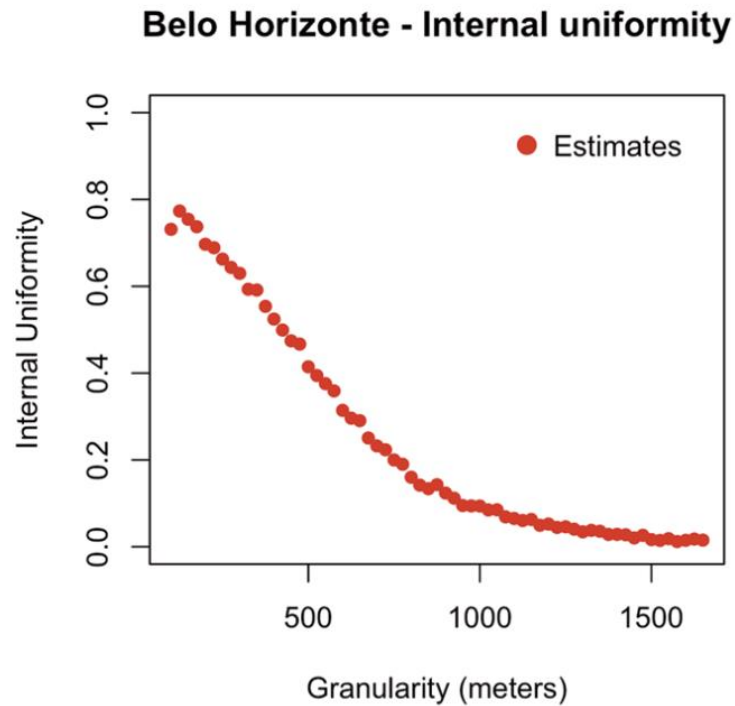
Ramos (2025)

What size areal unit should be selected when aggregating point data to create counts on a regular grid?



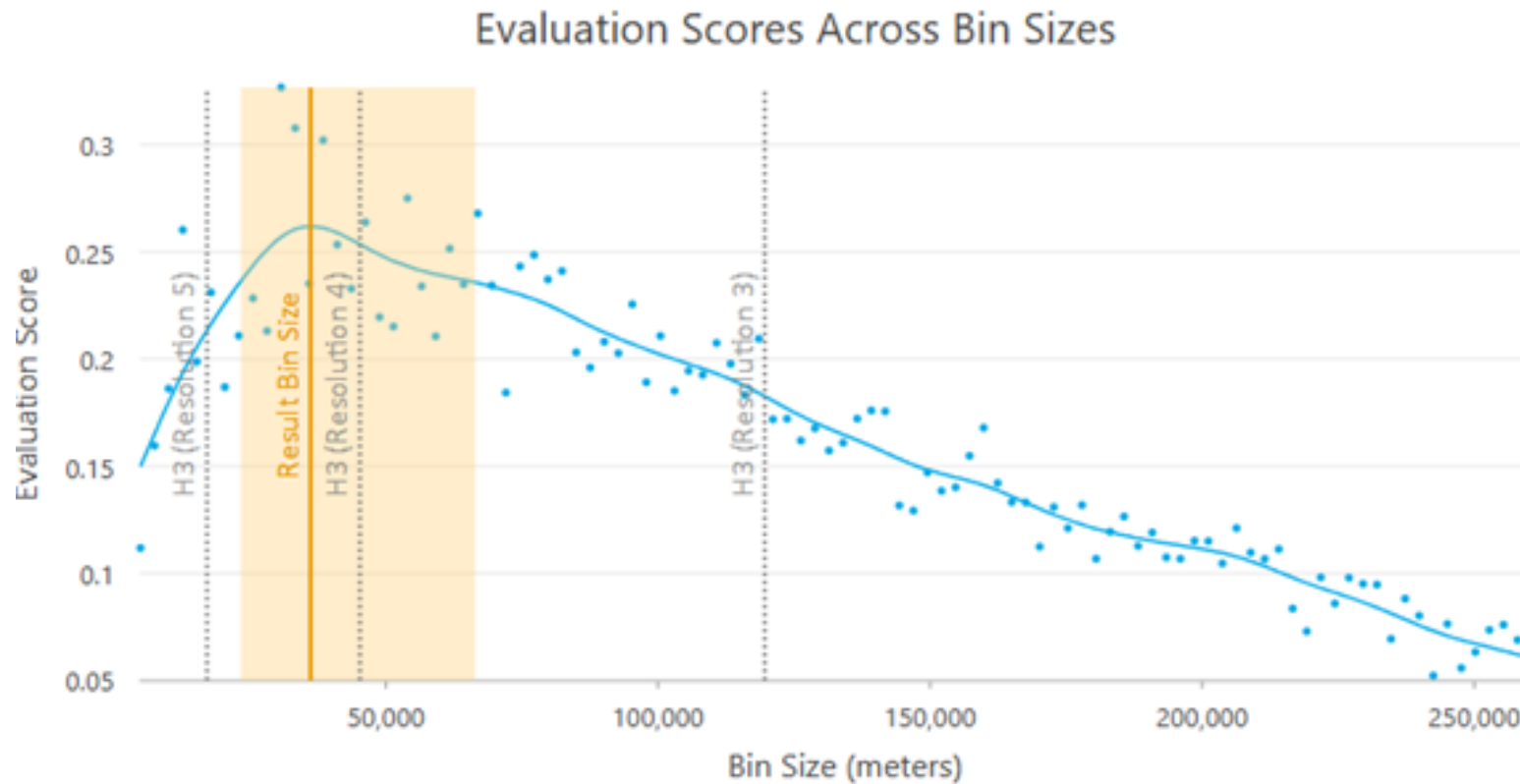
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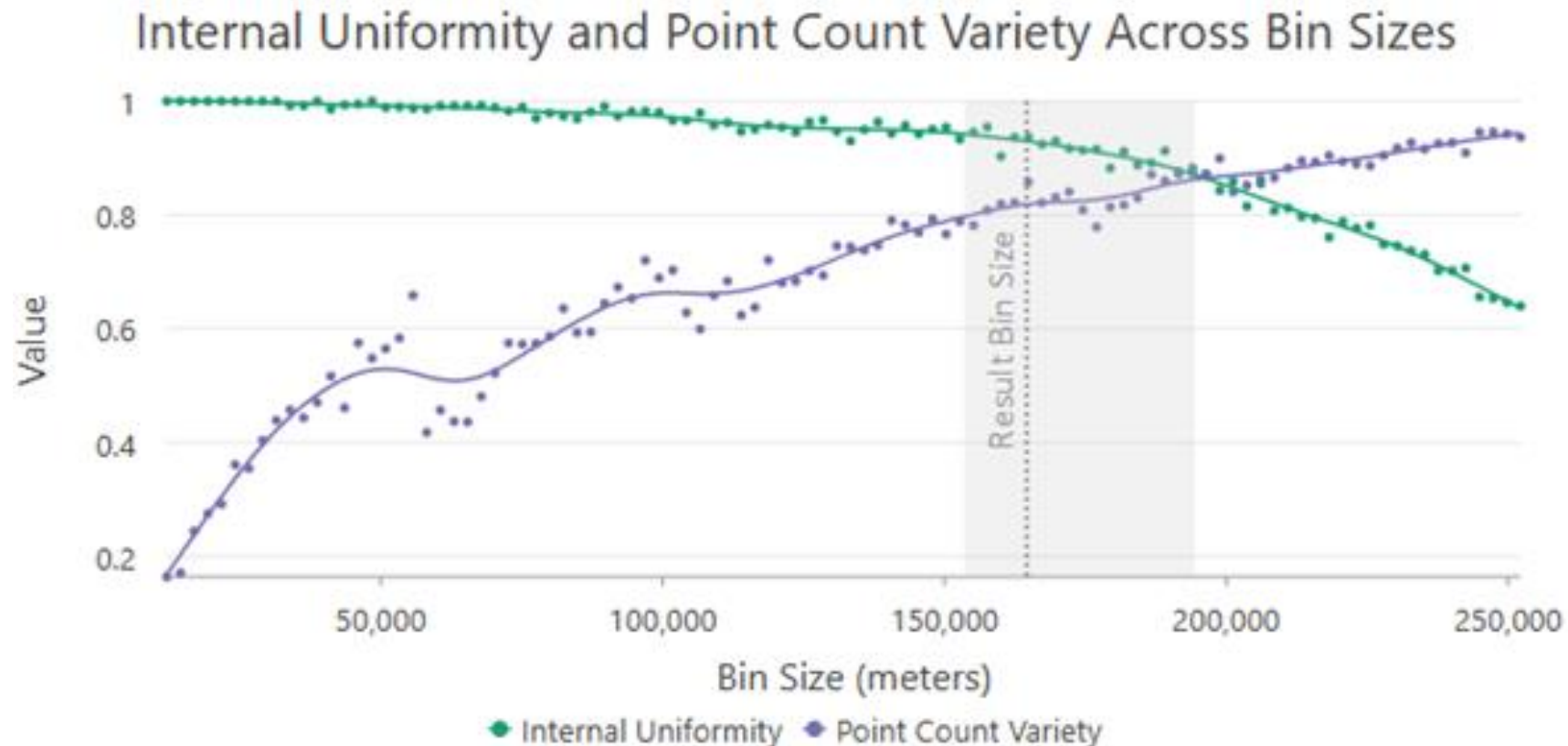
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STUDENTS PERSPECTIVE

How can GIScience justify hotspot detection, spatial autocorrelation, or neighborhood effects when the underlying spatial categories themselves may not be objectively real or fixed?

IDENTIFYING UNCERTAINTY

ACTIVITY

Purpose: To practice identifying sources of uncertainty in geographic analyses

We have a stack of suspicious spatial analysis cases. Most of these are simple cases. Today, we are putting them on trial. Each group will serve as an investigation team. Their responsibility, to diagnose each case, decide what uncertainty “charges” and “sub charges” apply and justify / explain how serious the problem is

ACTIVITY

Task:

- In groups of four, you will assign charges to each case. Use stickers for each category: pink for error, yellow for vagueness, green for ambiguity, and dark green for others.
- Each group will have to bring their case before the grand jury. We will select a pair of student from other groups who have not worked with the case before to serve as the judges.
- The jury will then have to make a verdict (approved, partially approved, over-ruled) based on the group's justification to see if the charges are appropriate.

CASE I

A city is mapping five-minute walking access to bus stops. One bus stop GPS point is displaced by 12 meters and appears inside a nearby river polygon instead of on the sidewalk. The analyst still uses the point to create walking-access buffers.

CASE II

A flood-risk model estimates the probability of flooding across a valley. Instead of showing probability, the final map converts the result into a single floodplain boundary: inside the line means at risk, outside means not at risk.

CASE IV

Researchers define “urban heat island” areas as places with surface temperatures above 95°F. Areas at 95.1°F are classified as heat islands, while nearby areas at 94.8°F are classified as not heat islands, even though the real temperature difference is tiny.

CASE III

During data entry, a residential parcel is accidentally coded as “industrial.” The parcel is then excluded from a housing suitability analysis because the model treats industrial land as unavailable for residential planning.

CASE V

A bridge was demolished three years ago, but it still appears in the transportation network dataset. A routing model uses the old bridge and concludes that residents have a short path to the nearest hospital.

CASE VI

A public health study examines food access by treating census tracts as neighborhoods. Everyone in the same tract is assigned the same neighborhood grocery access score, even though some residents live near stores and others live far away.

CASE VII

A river layer is digitized from an old map, but the line is heavily simplified to make the file smaller. Many small bends, side channels, and tributaries disappear. The layer is later used to estimate which parcels are near the river.

CASE VIII

Crime incidents are mapped as points and then aggregated into grid cells. When the analyst uses 250-meter cells, the hotspot appears near one intersection. When the analyst uses 1-kilometer cells, the hotspot shifts to a different part of the neighborhood.

BREAK

STUDENTS PERSPECTIVE

If some uncertainty is unknown, or even impossible to fully know, what is the practical value of discussing it?

In communicating geographic information to the public, does emphasizing uncertainty actually improve understanding, or does it simply function as a disclaimer: “this may be wrong, but we do not know exactly how or why”? How much uncertainty should researchers communicate in practice before it becomes confusing, unhelpful, or even damaging to public trust?

STUDENTS PERSPECTIVE

How, as scientists, can we change the narrative so that uncertainty is not viewed as a weakness, but rather as an opportunity to continue improving research, advancing findings, and strengthening the quality of data and results?

ACTIVITY

Purpose:

To make decisions when faced with uncertainty that impacts fitness for use, public investment, and disclosure

Scenario:

You are members of a planning review committee deciding where to prioritize public investment related to transit access in Cambridge and Somerville. Possible investments might include adding bus stops, improving pedestrian access to existing transit, strengthening connections to open space, or prioritizing underserved areas.

To support this decision, the city has outsourced the mapping and analysis work and received several maps from different analysts. All of the maps concern the same broad issue, but each map frames the problem differently.

You and your group members now need to convene for a stakeholder discussion based on the maps provided.

ACTIVITY

Task:

- You will work in groups of four.
- Each person in the group will take a different role in the stakeholder meeting that thinks about uncertainty in a different way.

Collectively, you should determine:

Whether each can map responsibly support the public planning decision AND create a statement about how the maps should be used in the planning decision.

You may annotate the maps as you wish. Please engage with concepts we have covered throughout the course on representation, scale, spatial relationships, etc.

MAP STATEMENTS

As the transit planner, I would use this map to ____ but I would not use it to ____ because ____.

As the GIS analyst, the biggest methodological uncertainty is ____ because ____.

As the community representative, I worry this map misses ____ because residents may experience access differently through ____.

As the public information officer, I would ask the team to disclose ____ before using this map for a public decision.

GROUP STATEMENT

Collectively, we think Map ____ is useful for ____, but it should not be used alone for _____. We would combine it with Map ____ because _____. The main uncertainty we would disclose is _____.

Before making a public investment decision, we would request _____.

STUDENTS PERSPECTIVE

If geospatial data are representations rather than reality itself, and if the external datasets used for validation can also carry their own measurement, classification, scale, or aggregation uncertainties, **what does validation actually validate?**

Does cross-validation reduce uncertainty, reveal disagreement among uncertain representations, or introduce another layer of uncertainty through the choice of reference data and evaluation criteria?

STUDENTS PERSPECTIVE

Thinking about the myths in this paper, I am adding the Gordian knot.

When complexity *and uncertainty overwhelm our ability to dissect a problem*, intuition from 'naive geography' might play *an important* role in forming an unconventional action that bypasses the tangled rules entirely.